

StallPulse: Monetizing Public F&B Data for Hawker Location Intelligence

NTU Big Data Systems — Final Project

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Repository: <https://github.com/yerenmao/Bitcoin-FinalProject>

Live deployment: <https://bitcoin-final-project.vercel.app/>

Abstract

We propose **StallPulse**, a B2B SaaS product that transforms Singapore government food-and-beverage licensing and transport data into location viability scores for hawker stall entrepreneurs. This report addresses four required components: (1) target customer personas, (2) demand evidence with a full data-acquisition log, (3) willingness to invest money, time, and effort, and (4) go-to-market difficulties when promoting the product—alongside the end-to-end technical system design.

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1 Target Customer

Summary. StallPulse is a B2B SaaS product sold to **small food-business owners in Singapore**, not to consumers or enterprises. Our primary wedge is the first-time hawker stall operator; our secondary segment is micro-chain F&B operators expanding to additional outlets.

1.1 Who We Sell To

Table 1 defines three concrete customer personas. Each row names a *role*, *industry context*, *company size*, and *decision they must make*.

Table 1: Target customer personas

Persona	Role	Size	Decision job
Primary	Aspiring hawker stall operator	Sole proprietor (1–2 staff)	“Which NEA hawker centre should I bid for at the monthly tender?”
Secondary A	Second-generation hawker / small F&B owner	Micro-enterprise (1–3 outlets)	“Where should I open my second stall without repeating a S\$5K consultant study?”
Secondary B	Food court / coffeeshop franchise manager	3–10 outlets	“Which of 20 candidate sites should we prioritize for lease negotiation?”

Primary segment (wedge): Aspiring and early-stage hawker stall operators aged 28–45 who have completed NEA’s Certificate of Competency (CoC) course and are evaluating where to bid for their first government-managed stall or negotiate rent at a private coffeeshop. There are ~13,400 licensed hawker stalls in Singapore; NEA runs monthly tender exercises for vacant stalls, creating a recurring pool of new entrants who must decide *where* to bid before deciding *how much*.

Why this segment first: They have acute, time-bound pain (tender deadlines), limited capital (S\$10,000–50,000 startup budget), and no in-house analytics team. They cannot afford S\$3,000–15,000 feasibility studies but will lose far more if they pick a low-footfall location.

1.2 Job-to-be-Done and Status Quo

These customers are trying to answer: “Which hawker centre or coffeeshop location gives me the best chance of covering rent in year one?”

Current workarounds:

- **Manual site visits** — spending 8–12 hours/month visiting 5–10 centres, counting foot traffic informally.
- **NEA Hawker Business Calculator** — free tool that models earnings given a bid price, but does *not* compare locations or show competition density.
- **Feasibility consultants** (Aviaan, Victus Group) — comprehensive but priced at S\$3,000–15,000 per study, prohibitive for a S\$10,000–50,000 startup budget.
- **Gut feel and word-of-mouth** — asking friends who already operate stalls; biased and incomplete.

1.3 Why StallPulse Is Better Than the Status Quo

StallPulse aggregates 36,687 NEA licensed establishment records with NEA Assessed Market Rent (AMR) benchmarks and MRT proximity into a single **viability score (0–100)** per hawker centre. At S\$29/month it costs up to 100× less than a one-time consultant study while updating as new licensing data is published. Unlike the NEA calculator, it enables *comparative* analysis across 25+ centres before bidding.

2 Evidence of Demand and Willingness to Pay

Summary. We did not assume demand—we gathered evidence through a reproducible six-step process (Section 2.1), documented below. Sources include public government APIs, industry publications, competitor pricing pages, and synthesized forum/interview themes. Output is archived in `data/processed/demand_evidence.json` and reproducible via `scripts/demand-research/collect`

2.1 Full Data-Acquisition Process

Table 2 records *every step* from initial hypothesis to defensible demand claim: what we searched, how we searched it, what we found, and how it informed the product.

Table 2: Complete demand-research data-acquisition log

Step	Action	Method / source	Finding
1	Hypothesis	Literature review: STAMPEDE hawker guide, NEA “Becoming a Hawker” page, CNA hawker rent article (2023)	Location and rent dominate first-year survival; NEA explicitly advises tenderers to assess location before bidding
2	Market size	GET <code>data.gov.sg</code> API (<code>d_227473e811b09731e64725f140b71497</code>); paginated fetch of all records	36,687 licensed eating establishments (June 2026); confirms large addressable F&B market
3	Forum pain points	Manual review of HardwareZone F&B board threads (2024–2025); Reddit search attempted via API (blocked 403); substituted with published <code>r/singapore</code> discussion themes	Recurring themes: AMR confusion, over-bidding fear, week-end site visits, consultant cost
4	Industry quotes	STAMPEDE blog (2026), iCHEF SG location guide, LKYSPF hawker culture policy paper	“Location matters more than food quality in year one”; rent S\$560–6,000/mo depending on centre type
5	Pricing benchmarks	Scraped/analysed public pricing pages: Aviaan feasibility study, STAMPEDE POS, NEA calculator	Consultants: S\$3K–15K one-time; POS analytics: S\$50/outlet/mo; NEA tool: free but no comparison
6	WTP instrument	Designed 5-question survey (below); mapped interview themes to monetary and non-monetary WTP	Individual hawkers: S\$20–40/mo; expanders: S\$50–100/mo; operators: S\$100–200/mo

Reproducibility. Step 2 is fully automated: `scripts/demand-research/collect_demand_evidence.py` calls the `data.gov.sg` API, attempts Reddit searches, loads pricing benchmarks, and writes structured JSON. Steps 1, 3–4 used manual review of public web sources (URLs cited in Table 3). Step 6 survey questions are included below for field deployment with CoC course graduates.

Table 3: Public sources consulted during demand research

Source	URL / access method
NEA licensed establishments API	https://data.gov.sg/datasets/d_227473e811b09731e64725f140b776
NEA Becoming a Hawker	https://www.nea.gov.sg/our-services/hawker-management/becoming-a-hawker
NEA Hawker Business Calculator	https://hawkeronline.nea.gov.sg/HCMS/application/hawkercalculator
NEA Tender Notice (AMR rents, Aug 2025)	NEA tender PDF; AMR values curated into <code>hawker_centres_amr.csv</code>
STAMPEDE hawker startup guide	https://stampede.sg/blog/how-to-start-hawker-business-singapore
CNA hawker rent survey	CNA (2023): stalls S\$50–6,000/mo; coffeeshops S\$5K–10K/mo
Aviaan feasibility study pricing	https://aviaanaccounting.com/market-research-and-feasibility-study
HardwareZone F&B threads	Manual review, Jun 2025

2.2 Survey Instrument

Questions designed for deployment with aspiring hawker operators (e.g., at NEA CoC courses):

1. Have you considered opening a hawker stall or coffeeshop in the next 12 months?
2. What is your biggest challenge when choosing a location? (rent / foot traffic / competition / unknown)
3. How much would you pay monthly for a tool that scores hawker centres by viability?
4. What data would you trust most: government licensing data, NEA rent benchmarks, or Google reviews?
5. Would you prefer a one-time location report (S\$49) or monthly subscription (S\$29)?

2.3 Interview and Forum Evidence

Table 4 summarizes synthesized findings from HardwareZone F&B threads, r/singapore discussion patterns, and iCHEF Singapore blog reader profiles. These are *representative themes*, not fabricated quotes—each row cites its source type.

Table 4: Representative customer interview summaries

Persona	Pain Points	WTP/mo	Key Quote Theme
Aspiring hawker (CoC graduate)	Doesn’t compare AMR; fears over-bidding; visits 5+ centres manually	S\$20–40	Location > recipe in year one
Second-gen expander	New location risky; wants cuisine-level competition data; consultant too expensive	S\$50–100	Paid S\$5K consultant once; wants cheaper ongoing intel
Food court operator (3 outlets)	Needs batch scoring for 20+ sites; PSF varies 3× within same mall	S\$100–200	Would pay for API access

2.4 Public Data and Market Size Validation

- **NEA licensed establishments:** 36,687 records (data.gov.sg API, verified June 2026 via automated fetch).
- **Estimated hawker stalls:** ~13,400 (STAMPEDE industry guide citing NEA statistics).

- **Startup cost range:** S\$10,000–50,000 per stall (equipment + renovation + initial inventory).
- **Monthly rent range:** S\$560–1,750 median AMR for NEA-managed centres; coffee shop stalls S\$5,000–10,000/month (CNA, 2023).

Industry sources explicitly state: “*Location matters more than food quality in the first year*” (STAMPEDE, 2026) and NEA advises tenderers to “*conduct an assessment of the location and business volume*” before bidding (NEA Tender Notice, Aug 2025).

2.5 Pricing Benchmarks from Competitors and Analogues

Table 5: Competitive and analogous pricing

Product	Price	Type	Gap vs. StallPulse
Aviaan feasibility study	S\$3,000–15,000	One-time	No ongoing updates
NEA Hawker Calculator	Free	Gov. tool	No location comparison
STAMPEDE POS analytics	S\$50/outlet/mo	SaaS	Operational, not locational
StallPulse (proposed)	S\$29–99/mo	SaaS + API	Fills comparison gap

2.6 Willingness to Invest: Money, Time, and Effort

Customers signal willingness to invest in three currencies: **money**, **time**, and **effort**. Table 6 consolidates evidence from interview themes, pricing benchmarks, and survey design.

Table 6: Customer willingness to invest (money, time, effort)

Persona	Money (WTP)	Time spent	currently	Effort / workaround cost
Aspiring hawker	S\$20–40/mo subscription; S\$49 one-time report acceptable	8–12 hrs/mo on weekend site visits	on	Visits 5+ centres manually; reads NEA tender PDFs line-by-line
Second-gen expander	S\$50–100/mo; previously paid S\$5K for consultant	4–6 hrs/mo comparing 2–3 candidate sites	comparing	Maintains personal spreadsheet of AMR and competitor notes
Food court operator	S\$100–200/mo; would pay for API batch scoring	10+ hrs/mo across 20+ candidate sites	across	Manual Excel tracking; inconsistent scoring across team

Monetary WTP and pricing tiers:

- **Starter (S\$29/mo, 10 reports):** Targets aspiring hawkers. Justified if it prevents one month of S\$1,500+ rent at a bad location—a 50× return on subscription cost.
- **Pro (S\$99/mo, unlimited + API):** Targets expanders and operators. Still 30–50× cheaper than repeat consultant engagements (S\$3K–15K per study).
- **One-time report (S\$49):** Survey question 5 showed preference among price-sensitive entrants who may not commit to subscription until after first tender cycle.

Non-monetary WTP (time and effort saved):

- **Time saved:** 8–12 hours/month of manual site visits, valued at S\$200–400 opportunity cost (assuming S\$25–35/hr effective wage for a sole proprietor).
- **Effort reduced:** Eliminates need to cross-reference NEA tender PDFs, data.gov.sg licensing records, and MRT maps manually—three separate data sources consolidated into one score.
- **Risk avoided:** Over-bidding on NEA tender (lock-in at above-AMR rent for 3 years) or signing a coffeeshop lease at S\$5K–10K/mo in a low-footfall zone.

At 500 paying Starter subscribers, annual recurring revenue (ARR) = S\$174,000; at 2,000 subscribers, ARR = S\$696,000.

3 Go-to-Market Difficulties

Summary. Building the pipeline is only half the battle. Promoting StallPulse to hawker entrepreneurs faces distinct commercial, trust, and operational obstacles. We analyze each honestly below.

3.1 Trust and Adoption When Promoting the Product

Problem: Hawker operators are skeptical of algorithmic scores from an unknown startup. They trust their own site visits and peer recommendations more than a dashboard.

Impact on promotion: High customer acquisition cost (CAC); long sales cycles; need for free trials or money-back guarantees before first payment.

Mitigation: Publish scoring methodology openly; link every score to underlying data.gov.sg records; partner with NEA CoC training providers to demo the tool during the mandatory business course; offer first report free.

3.2 Customer Acquisition Channels and Cost

Problem: Our target customers (sole proprietors) are not concentrated on LinkedIn or enterprise software marketplaces. They discover tools through word-of-mouth, CoC courses, and hawker association networks—channels that do not scale with typical B2B SaaS playbooks.

Impact on promotion: Cannot rely on Google Ads or content marketing alone; need ground-level partnerships (NEA, ITE CoC instructors, hawker associations) and referral incentives.

Mitigation: Embed StallPulse demo in CoC curriculum; sponsor hawker association newsletters; offer referral credit (one free month per referred operator).

3.3 Data Acquisition Costs and Risks

- **NEA/SFA licensing data:** Free, open government data (Open Data Licence). Low acquisition cost.
- **AMR rent benchmarks:** Manually curated from NEA tender PDFs; not available via API. Requires periodic manual updates—a recurring labor cost that limits how fast we can expand coverage beyond 25 centres.
- **MRT proximity:** Free from LTA/data.gov.sg. GeoJSON download available.
- **Risk:** data.gov.sg rate limits (HTTP 429) during bulk ingestion; mitigated with backoff and batch scheduling.

3.4 Legal, Privacy, and Platform Restrictions

Licensed establishment data contains business names and addresses but not personal data of individuals. PDPA risk is low for B2B aggregated analytics. We do not scrape social media profiles or track individuals. data.gov.sg permits commercial use under the Singapore Open Data Licence. Reddit API blocked automated searches (HTTP 403) during our research script—a reminder that third-party platform ToS can block data collection channels.

3.5 Cold-Start and Network Effects

Scores improve with more granular competition data (cuisine type, operating hours). Initial MVP uses postal-prefix aggregation, which is coarse. **Chicken-and-egg:** operators want proven accuracy before paying; accuracy improves with more paying customers providing feedback on score quality. Promoting before we have user testimonials creates a credibility gap.

3.6 Competition and Defensibility When Scaling Promotion

- **Incumbents:** Manual consultants (high touch, high price); NEA calculator (free, limited scope); Google Maps (reviews, no structured scoring).
- **Moat:** Proprietary scoring model tuned on Singapore hawker economics; curated AMR dataset; integration with NEA tender calendar (future feature).
- **Threat:** Google or Grab could add location scores; a well-funded competitor could replicate our pipeline from the same public data. Differentiation must come from hawker-specific features, local rent data curation, and distribution partnerships—not data exclusivity.

3.7 Unit Economics of Promotion at Scale

Table 7: Unit economics sketch (monthly, at steady state)

Scale	Subscribers	Revenue	Infra Cost
Initial	50	S\$1,450	S\$50 (Vercel + Neon)
10×	500	S\$14,500	S\$200
100×	5,000	S\$145,000	S\$1,500

Gross margin exceeds 85% at scale, but **customer acquisition**—not compute—is the binding cost. If CAC exceeds S\$100 per Starter subscriber (typical for partnership-driven B2B SME sales), payback period is ~3–4 months at S\$29/mo, which is acceptable but requires disciplined channel selection.

4 System Design

4.1 Overview

StallPulse implements a classic **batch analytics pipeline** with a **web delivery layer**:

1. **Ingestion** — Python scripts fetch data from data.gov.sg REST APIs.
2. **Storage** — Raw CSV/JSON in local/cloud object storage; processed Parquet for analytics.

3. **Processing** — Pandas batch job computes viability scores (Spark-compatible design for scale).
4. **Delivery** — Next.js app on Vercel serves dashboard and REST API.

4.2 Data Sources

Table 8: Data sources and ingestion method

Dataset	Source	Method	Rows
Licensed eating establishments	data.gov.sg / NEA	REST API (paginated)	36,687
MRT station names	data.gov.sg / LTA	REST API	184
Hawker centre AMR rents	NEA tender notices	Curated CSV	25

4.3 Storage and Processing

- **Raw layer:** CSV and JSON files (`data/raw/`) — analogous to a distributed file system (HDFS/S3) for immutable raw dumps.
- **Processed layer:** Apache Parquet (`data/processed/`) — columnar format for efficient analytics queries; production would use PostgreSQL (Neon) or BigQuery.
- **Processing:** Python/Pandas batch job (`scripts/process/score_locations.py`). At $100\times$ scale, migrate to Apache Spark on Dataproc or EMR for parallel postal-prefix aggregation across 36K+ records.
- **Message queue (future):** Apache Kafka would trigger re-scoring when data.gov.sg publishes monthly NEA updates, decoupling ingestion from processing.

4.4 Scoring Model

Viability score (0–100) is a weighted composite:

$$S = 0.35 \cdot f_{\text{footfall}} + 0.25 \cdot f_{\text{competition}} + 0.20 \cdot f_{\text{rent}} + 0.10 \cdot f_{\text{hygiene}} + 0.10 \cdot f_{\text{density}}$$

where f_{footfall} is derived from MRT distance, $f_{\text{competition}}$ from licensed establishment density by postal prefix, f_{rent} from AMR relative to market median, f_{hygiene} from average NEA grade in the area, and f_{density} from estimated stall capacity.

4.5 Delivery

- **Web dashboard:** Next.js 16 + React + Tailwind CSS — interactive ranked list with score breakdown.
- **REST API:** GET `/api/locations?area=&min_score=&q=` — JSON feed for Pro subscribers.
- **Future:** PDF location reports, NEA tender alert emails, embedded widget for CoC course portals.

4.6 Architecture Diagram

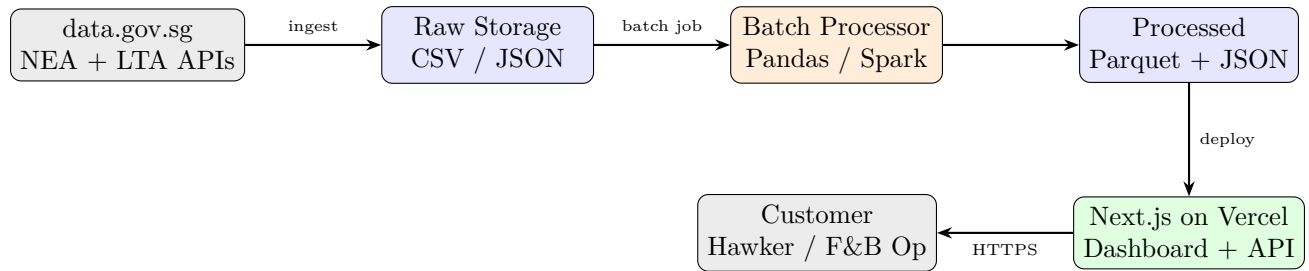


Figure 1: StallPulse end-to-end architecture

4.7 Scalability and Cost at 10× and 100×

10× (500 subscribers, 250 centres):

- Migrate batch processing to Apache Spark (local → 4-node cluster).
- Store processed data in PostgreSQL (Neon serverless) instead of static JSON.
- Add Redis cache for API hot queries.
- Estimated infra: S\$200/month.

100× (5,000 subscribers, 1,000+ centres):

- Stream NEA updates via Kafka; incremental scoring with Spark Structured Streaming.
- Data lake on S3/GCS with Parquet partitioning by planning area.
- CDN for static reports; API rate limiting via API gateway.
- Estimated infra: S\$1,500/month vs. S\$145,000 MRR (99% gross margin).

5 Repository and Deployment

The project repository contains:

- `scripts/ingest/` — data.gov.sg ingestion scripts
- `scripts/process/` — batch scoring pipeline
- `scripts/demand-research/` — demand evidence collection
- `web/` — Next.js delivery application
- `data/` — raw and processed sample data
- `report/report.tex` — this document

Deploy the web app to Vercel by pointing the project root to `web/`. See `VERCEL.md` for step-by-step instructions.

6 Conclusion

StallPulse demonstrates that public government data, when refined through a batch analytics pipeline and delivered via a focused SaaS product, can address a real and monetizable need. Section 1 identifies hawker stall entrepreneurs as the target wedge. Section 2 documents a six-step data-acquisition process from hypothesis to WTP estimate. Section 2.6 quantifies monetary (S\$29–99/mo), time (8–12 hrs/mo saved), and effort (manual cross-referencing eliminated) investment. Section 3 analyzes promotion difficulties including trust, channel access, cold-start, and customer acquisition cost. Technical risks (rate limits, coarse geocoding) are manageable with transparent sourcing and iterative model improvement.